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United States Patent Application Publication	20250285996
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Ni; Pei-Rong et al.

SEMICONDUCTOR STRUCTURE

Abstract

A semiconductor structure including an interposer is provided. The interposer includes a substrate, a redistribution layer (RDL), a first integrated passive device (IPD), and a second IPD. The RDL is located on the substrate. The first IPD is located in the substrate. The second IPD is located in the RDL.

Inventors: Ni; Pei-Rong (Hsinchu, TW), Lu; Chun-Lin (Hsinchu City, TW), Chang; Shou-Zen (Hsinchu City, TW)

Applicant: Powerchip Semiconductor Manufacturing Corporation (Hsinchu, TW)

Family ID: 1000007797570

Assignee: Powerchip Semiconductor Manufacturing Corporation (Hsinchu, TW)

Appl. No.: 18/620984

Filed: March 28, 2024

Foreign Application Priority Data

TW	113108237	Mar. 06, 2024
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Publication Classification

Int. Cl.: H01L23/64 (20060101); H01L23/00 (20060101); H01L23/538 (20060101)

U.S. Cl.:

CPC H01L23/642 (20130101); H01L23/5383 (20130101); H01L23/5384 (20130101); H01L24/16 (20130101); H01L2224/16227 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113108237, filed on Mar. 6, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of specification.

BACKGROUND

Technical Field

[0002] The invention relates to a semiconductor structure, and particularly relates to a semiconductor structure including an integrated passive device (IPD).

Description of Related Art

[0003] Currently, the IPD is widely used in the semiconductor structure, and the IPD is mounted on the interposer and next to the main die. Since the IPD occupies the area on the interposer used for mounting the die, the degree of freedom and the flexibility of the package design will be reduced.

SUMMARY

[0004] The invention provides a semiconductor structure, which can improve the degree of freedom and the flexibility of the package design.

[0005] The invention provides a semiconductor structure, which includes an interposer. The interposer includes a substrate, a redistribution layer (RDL), a first integrated passive device (IPD), and a second IPD. The RDL is located on the substrate. The first IPD is located in the substrate. The second IPD is located in the RDL.

[0006] According to an embodiment of the invention, in the semiconductor structure, the first IPD and the second IPD may be electrically connected to each other.

[0007] According to an embodiment of the invention, in the semiconductor structure, the first IPD and the second IPD may be connected in series.

[0008] According to an embodiment of the invention, in the semiconductor structure, the first IPD and the second IPD may be connected in parallel.

[0009] According to an embodiment of the invention, in the semiconductor structure, the first IPD and the second IPD may be electrically insulated from each other.

[0010] According to an embodiment of the invention, in the semiconductor structure, the first IPD and the second IPD may be separated from each other.

[0011] According to an embodiment of the invention, in the semiconductor structure, the second IPD may overlap the first IPD.

[0012] According to an embodiment of the invention, in the semiconductor structure, the second IPD does not overlap the first IPD.

[0013] According to an embodiment of the invention, in the semiconductor structure, the first IPD may be a capacitor.

[0014] According to an embodiment of the invention, in the semiconductor structure, the second IPD may be a capacitor.

[0015] According to an embodiment of the invention, in the semiconductor structure, the RDL may include a dielectric layer, an interconnect structure, and a pad. The dielectric layer is located on the substrate. The interconnect structure is located in the dielectric layer. The pad is located above the interconnect structure and is located in the dielectric layer.

[0016] According to an embodiment of the invention, in the semiconductor structure, the first IPD may be electrically connected to the interconnect structure.

[0017] According to an embodiment of the invention, in the semiconductor structure, the second IPD may be electrically connected to the interconnect structure.

[0018] According to an embodiment of the invention, in the semiconductor structure, the second

IPD may be located in the dielectric layer.

[0019] According to an embodiment of the invention, in the semiconductor structure, the dielectric layer may include a first surface and a second surface opposite to each other. The first surface may be adjacent to the substrate.

[0020] According to an embodiment of the invention, in the semiconductor structure, the first IPD and the second IPD may be adjacent to the first surface. The pad may be adjacent to the second surface.

[0021] According to an embodiment of the invention, the semiconductor structure may further include a die. The die is located on the second surface. The die may be electrically connected to the pad.

[0022] According to an embodiment of the invention, the semiconductor structure may further include a connection terminal. The connection terminal is located between the die and the pad.

[0023] According to an embodiment of the invention, in the semiconductor structure, the interposer may further include a through-substrate via (TSV). The TSV passes through the substrate.

[0024] According to an embodiment of the invention, in the semiconductor structure, the TSV may extend into the dielectric layer.

[0025] Based on the above description, in the semiconductor structure according to the invention, the first IPD is located in the substrate, and the second IPD is located in the RDL. Therefore, the first IPD and the second IPD do not occupy the area on the interposer used for mounting the die. In this way, the degree of freedom and the flexibility of the package design can be improved, and the difficulty and the cost of the back end package process can be reduced.

[0026] In order to make the aforementioned and other objects, features and advantages of the invention comprehensible, several exemplary embodiments accompanied with drawings are described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

[0028] FIG. 1 is a cross-sectional view of a semiconductor structure according to some embodiments of the invention.

[0029] FIG. 2 is a cross-sectional view of a semiconductor structure according to other embodiments of the invention.

[0030] FIG. 3 is a cross-sectional view of a semiconductor structure according to other embodiments of the invention.

DESCRIPTION OF THE EMBODIMENTS

[0031] The embodiments are described in detail below with reference to the accompanying drawings, but the embodiments are not intended to limit the scope of the invention. For the sake of easy understanding, the same components in the following description will be denoted by the same reference symbols. In addition, the drawings are for illustrative purposes only and are not drawn to the original dimensions. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0032] FIG. 1 is a cross-sectional view of a semiconductor structure according to some embodiments of the invention. FIG. 2 is a cross-sectional view of a semiconductor structure according to other embodiments of the invention. FIG. 3 is a cross-sectional view of a semiconductor structure according to other embodiments of the invention.

[0033] Referring to FIG. 1A to FIG. 3, a semiconductor structure **100** includes an interposer **102**. In some embodiments, the interposer **102** may be a silicon interposer (Si-interposer). The interposer **102** includes a substrate **104**, a redistribution layer (RDL) **106**, an integrated passive device (IPD) **108**, and an IPD **110**. In some embodiments, the substrate **104** may be a semiconductor substrate such as a silicon substrate.

[0034] The RDL **106** is located on the substrate **104**. The RDL **106** may include a dielectric layer **112**, an interconnect structure **114**, and a pad **116**. The dielectric layer **112** is located on the substrate **104**. The dielectric layer **112** may include a first surface S1 and a second surface S2 opposite to each other. The first surface S1 may be adjacent to the substrate **104**. In some embodiments, the dielectric layer **112** may be a multilayer structure. In some embodiments, the material of the dielectric layer **112** is, for example, silicon oxide, silicon nitride, or a combination thereof. The interconnect structure **114** is located in the dielectric layer **112**. The interconnect structure **114** may include a conductive line **114a**, a plug **114b**, or a combination thereof. In some embodiments, the plug **114b** may be a contact plug or a via plug. In some embodiments, the material of the interconnect structure **114** is, for example, copper, aluminum, tungsten, tantalum, tantalum nitride, titanium, titanium nitride, or a combination thereof. The pad **116** is located above the interconnect structure **114** and is located in the dielectric layer **112**. The pad **116** may be adjacent to the second surface S2. The pad **116** may be electrically connected to the corresponding interconnect structure **114**. The material of the pad **116** is, for example, a conductive material such as metal (e.g., copper, aluminum, or tungsten).

[0035] The IPD **108** is located in the substrate **104**. That is, IPD **108** may be embedded in the substrate **104**. The IPD **108** may be adjacent to the first surface S1. The IPD **108** may be electrically connected to the interconnect structure **114**. In some embodiments, the IPD **108** may be a capacitor. In some embodiments, the IPD **108** is, for example, a trench capacitor.

[0036] The IPD **110** is located in the RDL **106**. That is, the IPD **110** may be embedded in the RDL **106**. Since the IPD **108** is located in the substrate **104**, and the IPD **110** is located in the RDL **106**, the IPD **108** and the IPD **110** do not occupy the area on the interposer **102** used for mounting the die. In some embodiments, the IPD **110** may be located in the dielectric layer **112**. The IPD **110** may be adjacent to the first surface S1. The IPD **110** may be electrically connected to the interconnect structure **114**. The IPD **108** and the IPD **110** may be separated from each other. In some embodiments, the IPD **108** and the IPD **110** may be separated from each other by the dielectric layer **112**. In some embodiments, the IPD **110** may be a capacitor. In some embodiments, the IPD **110** is, for example, a trench capacitor.

[0037] In some embodiments, as shown in FIG. 1, the IPD **108A** and the IPD **110A** may be electrically connected to each other, but the invention is not limited thereto. In other embodiments, as shown in FIG. 2 and FIG. 3, the IPD **108A** and the IPD **110A** may be electrically insulated from each other. In some embodiments, as shown in FIG. 1, the IPD **108A** and the IPD **110A** may be electrically connected to each other by the interconnect structure **114**. In some embodiments, the IPD **108A** and the IPD **110A** may be connected in series. In some embodiments, the IPD **108A** and the IPD **110A** may be connected in parallel. In some embodiments, as shown in FIG. 1, the IPD **108A** and the IPD **110A** may be connected in series or in parallel by the interconnect structure **114**.

[0038] In some embodiments, as shown in FIG. 1 to FIG. 3, the IPD **108B** and the IPD **110B** may be electrically insulated from each other, but the invention is not limited thereto. In other embodiments, although not shown in the figure, the IPD **108B** and the IPD **110B** may be electrically connected to each other.

[0039] In some embodiments, as shown in FIG. 1, the IPD **110A** may overlap the IPD **108A**, and the IPD **110B** may overlap the IPD **108B**. In other embodiments, as shown in FIG. 2, the IPD **110A** does not overlap the IPD **108A**, and the IPD **110B** may overlap the IPD **108B**. In other embodiments, as shown in FIG. 3, the IPD **110A** does not overlap the IPD **108A**, and the IPD **110B** does not overlap the IPD **108B**.

[0040] The interposer **102** may further include a through-substrate via (TSV) **118**. The TSV **118** passes through the substrate **104**. The TSV **118** may extend into the dielectric layer **112**. The TSV **118** may be electrically connected to the corresponding interconnect structure **114**. In some embodiments, the material of the TSV **118** is, for example, copper, tantalum, tantalum nitride, or a combination thereof.

[0041] The semiconductor structure **100** may further include a die **120**. The die **120** is located on the second surface S2. The die **120** may be electrically connected to the pad **116**. In some embodiments, the die **120** may be a high bandwidth memory (HBM) die or a system on chip (SOC) die. When the number of the dies **120** is plural, the dies **120** may include HBM dies, SOC dies, or combinations thereof.

[0042] The semiconductor structure **100** may further include a connection terminal **122**. The connection terminal **122** is located between the die **120** and the pad **116**. The die **120** may be electrically connected to the pad **116** by the connection terminal **122**. In some embodiments, the IPD **108** may be electrically connected to the corresponding die **120** by the corresponding interconnect structure **114**, the corresponding pad **116**, and the corresponding connection terminal **122**. In some embodiments, the IPD **110** may be electrically connected to the corresponding die **120** by the corresponding interconnect structure **114**, the corresponding pad **116**, and the corresponding connection terminal **122**. The connection terminal **122** may be a bump (e.g., solder ball), but the invention is not limited thereto.

[0043] Based on the above embodiments, in the semiconductor structure **100**, the IPD **108** is located in the substrate **104**, and the IPD **110** is located in the RDL **106**. Therefore, the IPDs do not occupy the area on the interposer **102** used for mounting the die **120**. In this way, the degree of freedom and the flexibility of the package design can be improved, and the difficulty and the cost of the back end package process can be reduced.

[0044] In summary, in the semiconductor structure of the aforementioned embodiments, since the IPD does not occupy the area on the interposer used for mounting the die, the degree of freedom and the flexibility of the package design can be improved, and the difficulty and the cost of the back end package process can be reduced.

[0045] Although the invention has been described with reference to the above embodiments, it will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit of the invention. Accordingly, the scope of the invention is defined by the attached claims not by the above detailed descriptions.

Claims

1. A semiconductor structure, comprising: an interposer comprising: a substrate; a redistribution layer (RDL) located on the substrate; a first integrated passive device (IPD) located in the substrate; and a second IPD located in the RDL.
2. The semiconductor structure according to claim 1, wherein the first IPD and the second IPD are electrically connected to each other.
3. The semiconductor structure according to claim 2, wherein the first IPD and the second IPD are connected in series.
4. The semiconductor structure according to claim 2, wherein the first IPD and the second IPD are connected in parallel.
5. The semiconductor structure according to claim 1, wherein the first IPD and the second IPD are electrically insulated from each other.
6. The semiconductor structure according to claim 1, wherein the first IPD and the second IPD are separated from each other.
7. The semiconductor structure according to claim 1, wherein the second IPD overlaps the first IPD.

- 8.** The semiconductor structure according to claim 1, wherein the second IPD does not overlap the first IPD.
- 9.** The semiconductor structure according to claim 1, wherein the first IPD comprises a capacitor.
- 10.** The semiconductor structure according to claim 1, wherein the second IPD comprises a capacitor.
- 11.** The semiconductor structure according to claim 1, wherein the RDL comprises: a dielectric layer located on the substrate; an interconnect structure located in the dielectric layer; and a pad located above the interconnect structure and located in the dielectric layer.
- 12.** The semiconductor structure according to claim 11, wherein the first IPD is electrically connected to the interconnect structure.
- 13.** The semiconductor structure according to claim 11, wherein the second IPD is electrically connected to the interconnect structure.
- 14.** The semiconductor structure according to claim 11, wherein the second IPD is located in the dielectric layer.
- 15.** The semiconductor structure according to claim 11, wherein the dielectric layer comprises a first surface and a second surface opposite to each other, and the first surface is adjacent to the substrate.
- 16.** The semiconductor structure according to claim 15, wherein the first IPD and the second IPD are adjacent to the first surface, and the pad is adjacent to the second surface.
- 17.** The semiconductor structure according to claim 15, further comprising: a die located on the second surface and electrically connected to the pad.
- 18.** The semiconductor structure according to claim 17, further comprising: a connection terminal located between the die and the pad.
- 19.** The semiconductor structure according to claim 11, wherein the interposer further comprises: a through-substrate via (TSV) passing through the substrate.
- 20.** The semiconductor structure according to claim 19, wherein the TSV extends into the dielectric layer.
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